

1. What is Statistics?

Statistics is a branch of mathematics that deals with:

- collecting data
- analyzing data
- interpreting data
- presenting data

Statistics helps us **understand patterns, trends, and variability** in data.

2. Measures of Central Tendency

These measures describe the **center** of a dataset.

◆ Mean (Average)

The **mean** is the sum of all values divided by the number of values.

Formula:

$$\text{Mean} = \frac{\sum x}{n} \quad \text{Mean} = \frac{\sum x}{n}$$

Example:

Data: 2, 4, 6

$$\text{Mean} = (2 + 4 + 6) / 3 = 4$$

Key points:

- Sensitive to outliers
- Commonly used in ML and data analysis

◆ Median

The **median** is the middle value when data is sorted.

Example:

Data: 1, 3, 5 → Median = 3

Data: 1, 3, 5, 7 → Median = $(3 + 5) / 2 = 4$

Key points:

- Robust to outliers
- Often preferred for skewed data (e.g., income)

◆ Mode

The **mode** is the value that appears most frequently.

Example:

Data: 1, 2, 2, 3 → Mode = 2

Key points:

- Can be more than one mode
- Useful for categorical data

◆ Mean vs Median vs Mode

- **Mean:** affected by extreme values
- **Median:** resistant to outliers
- **Mode:** shows most common value

3. Measures of Variability

These describe how **spread out** the data is.

◆ Variance

Variance measures how far data points are from the mean.

Formula (population):

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N} \quad \sigma^2 = \frac{1}{N} \sum (x - \mu)^2$$

Key points:

- Higher variance → more spread
- Units are squared (not intuitive)

◆ Standard Deviation

The **standard deviation** is the square root of variance.

Formula:

$$\sigma = \sqrt{\sigma^2} \quad \sigma = \sqrt{\frac{1}{N} \sum (x - \mu)^2}$$

Key points:

- Same units as data
- Shows typical distance from the mean
- Widely used in ML and statistics

4. Distributions

A **distribution** describes how values are spread across possible outcomes.

Common types:

- Normal distribution
- Exponential distribution
- Uniform distribution

◆ Normal Distribution (Gaussian)

A **bell-shaped**, symmetric distribution.

Characteristics:

- Mean = Median = Mode
- Symmetrical around the mean
- Defined by mean (μ) and standard deviation (σ)

68–95–99.7 rule:

- 68% within 1σ
- 95% within 2σ
- 99.7% within 3σ

Used in:

- Natural phenomena
- Measurement errors
- Many ML assumptions

◆ Exponential Distribution

Models **time between events**.

Characteristics:

- Right-skewed

- No upper limit
- Memoryless property

Used in:

- Waiting times
- Failure rates
- Queue systems

5. Histograms

A **histogram** is a graphical representation of data distribution.

How it works:

- Data is grouped into bins
- Height of bar = frequency

Why histograms matter:

- Show shape of distribution
 - Identify skewness
 - Detect outliers
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6. Population vs Sample

◆ Population

The entire group of interest.

◆ Sample

A subset of the population used for analysis.

Why sampling matters:

- Real populations are often too large
 - Samples help estimate population parameters
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◆ Parameters vs Estimates

- **Parameter** → true population value
 - **Estimate** → calculated from sample
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7. What is a Mathematical Model?

A **mathematical model** is a simplified representation of reality using numbers and equations.

In ML:

- Models learn patterns from data
 - Statistics provides the foundation
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8. Why Statistics Matters in Machine Learning

- Understanding data distributions
- Feature scaling
- Model assumptions
- Error analysis
- Performance evaluation